

A BLOCK NAKANO EXAMPLE ANSWERING PROBLEM 4.8 OF SMITH–TROYANSKI

LANE 19 DISCOVERY PACKET

SOURCE AND STATUS

The source paper is Richard J. Smith and Stanimir Troyanski, *Approximation of norms on Banach spaces*, arXiv:1804.05660. Problem 4.8 asks whether there exists a nondecreasing sequence $(p_k)_{k=1}^\infty$, with $p_1 \geq 1$ and $p_k \rightarrow \infty$, such that one of two stated test suprema is infinite in the symmetric Nakano space $h_{(p_k)}^S(\Gamma)$.

The result below answers the problem affirmatively by making the first supremum infinite.

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RICHARD J. SMITH AND STANIMIR TROYANSKI

Problem 4.8. *Does there exist a non-decreasing sequence $(p_k)_{k=1}^\infty$, with $p_1 \geq 1$, $p_k \rightarrow \infty$, such that*

$$\sup \left\{ \left\| \sum_{k=1}^n \frac{e_{\gamma_k}}{\lambda(k)} \right\| : n \in \mathbb{N} \right\} = \infty,$$

or

$$\sup \left\{ \left\| \sum_{k=1}^n (\mu(k) - \mu(k-1)) e_{\gamma_k} \right\| : n \in \mathbb{N} \right\} = \infty,$$

with respect to $h_{(p_n)}^S(\Gamma)$?

For some classes of spaces, for example, Orlicz spaces whose standard basis is shrinking, the implication in Corollary 4.5 is reversible – see Theorem 5.3 below. We close the section

FIGURE 1. Problem 4.8 in the source paper, page 20.

DEFINITIONS FROM THE SOURCE

Let Γ be an infinite set and let (p_k) be a nondecreasing sequence with $p_1 \geq 1$. The symmetric Nakano space $h_{(p_k)}^S(\Gamma)$ consists of the real functions x on Γ such that

$$\phi(x/\rho) < \infty \quad (\rho > 0),$$

where

$$\phi(x) = \sup \left\{ \sum_{k=1}^{\infty} |x(\gamma_k)|^{p_k} : (\gamma_k)_{k=1}^{\infty} \text{ is a sequence of distinct points of } \Gamma \right\}.$$

Its norm is

$$\|x\| = \inf \{ \rho > 0 : \phi(x/\rho) \leq 1 \}.$$

For a fixed sequence of distinct points (γ_k) , the source defines

$$\lambda(n) = \left\| \sum_{k=1}^n e_{\gamma_k} \right\|.$$

PROOF INTUITION

The construction makes p_k constant on extremely long blocks. On the block where $p_k = m$, the defining equation for $\lambda(k)$ contains a current-block term of the form $r\lambda(k)^{-m}$, where r is the position inside that block. Once r is large compared with all previous blocks, the previous-block contribution is at most one half, so

$$\lambda(k)^{-m} \geq \frac{1}{2r}.$$

The vector from Problem 4.8 then has a modular contribution bounded below by a harmonic sum over that block. By making the block length enormous, this harmonic sum beats the trial radius m , forcing the corresponding partial-sum norm to exceed m . This happens for arbitrarily large m .

Theorem 1. *For every infinite set Γ there is a nondecreasing sequence $(p_k)_{k=1}^\infty$ of positive integers with $p_k \rightarrow \infty$ such that, in $h_{(p_k)}^S(\Gamma)$,*

$$\sup_{n \in \mathbb{N}} \left\| \sum_{k=1}^n \frac{e_{\gamma_k}}{\lambda(k)} \right\| = \infty$$

for every sequence (γ_k) of distinct points of Γ . Consequently Problem 4.8 of the source paper has an affirmative answer.

Proof. Choose integers

$$0 = N_0 < N_1 < N_2 < \cdots$$

recursively as follows. Put $L_m = N_m - N_{m-1}$ and

$$R_m = \max \{1, \lceil (2N_{m-1})^m \rceil\}.$$

At stage m , choose L_m so large that

$$L_m \geq R_m \exp(4m^m).$$

This is possible because N_{m-1} has already been chosen. Define

$$p_k = m \quad \text{whenever} \quad N_{m-1} < k \leq N_m.$$

Then (p_k) is nondecreasing, $p_1 \geq 1$, and $p_k \rightarrow \infty$.

Fix a sequence of distinct points (γ_k) in Γ . Since all coefficients of $\sum_{i=1}^k e_{\gamma_i}$ are equal to 1, and the exponents are nondecreasing, the Nakano modular is maximized by placing these nonzero coordinates in the first k positions. Therefore $\lambda(k)$ is the unique number at least 1 satisfying

$$(1) \quad \sum_{i=1}^k \lambda(k)^{-p_i} = 1.$$

Now fix m and take $k = N_{m-1} + r$ with $R_m \leq r \leq L_m$. For these k , equation (1) gives

$$1 = \sum_{i=1}^{N_{m-1}} \lambda(k)^{-p_i} + r\lambda(k)^{-m}.$$

The inequality $r\lambda(k)^{-m} \leq 1$ implies $\lambda(k) \geq r^{1/m} \geq 2N_{m-1}$. Since every earlier exponent is at least 1,

$$\sum_{i=1}^{N_{m-1}} \lambda(k)^{-p_i} \leq N_{m-1} \lambda(k)^{-1} \leq \frac{1}{2}.$$

Consequently

$$r\lambda(k)^{-m} \geq \frac{1}{2}, \quad \text{hence} \quad \lambda(k)^{-m} \geq \frac{1}{2r}.$$

Let

$$x_m = \sum_{k=1}^{N_m} \frac{e_{\gamma_k}}{\lambda(k)}.$$

Using the particular ordering $(\gamma_1, \dots, \gamma_{N_m})$ in the defining supremum for ϕ , we obtain

$$\phi(x_m/m) \geq \sum_{k=N_{m-1}+R_m}^{N_m} \left(\frac{1}{m\lambda(k)} \right)^m \geq \frac{1}{2m^m} \sum_{r=R_m}^{L_m} \frac{1}{r}.$$

The elementary integral estimate $\sum_{r=R}^L r^{-1} \geq \log((L+1)/R)$ gives

$$\phi(x_m/m) \geq \frac{1}{2m^m} \log \frac{L_m+1}{R_m} \geq \frac{1}{2m^m} \log \frac{L_m}{R_m} \geq 2.$$

Thus $\phi(x_m/m) > 1$, so $\|x_m\| > m$. Since this holds for every m , the sequence of partial-sum norms is unbounded. $\square$

VERIFICATION AND NOVELTY CHECK

The proof uses only the source definition of $h_{(p_k)}^S(\Gamma)$ and of $\lambda(k)$ from Section 4 of arXiv:1804.05660. The key inequalities are the current-block domination estimate and the harmonic lower bound above.

Bounded novelty search was performed before promotion using exact and close phrases including “Does there exist a non-decreasing sequence”, “h^S”, “lambda(k)”, “symmetric Nakano”, and “Approximation of norms on Banach spaces”. No direct later answer to Problem 4.8 was found in that quick search.

REFERENCES

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- [1] R. J. Smith and S. Troyanski, *Approximation of norms on Banach spaces*, arXiv:1804.05660.